

Cyprus AI — Microsoft Teams Integration Guide

Step-by-step instructions to make the policy chatbot available to employees in Microsoft Teams

Overview

Cyprus AI is a FastAPI web service that exposes a bot endpoint at `/api/messages` (see `app/bot/bot_handler.py`). To let employees add the bot to their Teams chat, we need to register it in Azure, create an Azure Bot resource, deploy the service behind a public HTTPS endpoint, package a Teams app, and publish that app to the tenant. This guide walks through every step.

Prerequisites: an account with permission to create resources in the company's Microsoft 365 / Entra tenant (a normal employee account is not enough), and the Ubuntu server where the app is deployed.

Step 1 — Register a Bot App in Azure AD (Entra ID)

- 1 Go to <https://portal.azure.com> and sign in with an administrative account.
- 2 Navigate to **Microsoft Entra ID** → **App registrations** → **+ New registration**.
- 3 Set the **Name** to **Cyprus AI Bot**.
- 4 Under **Supported account types**, choose **Accounts in this organizational directory only** (single tenant), since this is for your company only.
- 5 Click **Register**.

From the Overview page, copy these two values (you will need them later):

Field	Copy from	Maps to .env
Application (client) ID	Overview page	MicrosoftAppId
Directory (tenant) ID	Overview page	MicrosoftAppTenantId

- 1 In the left menu, open **Certificates & secrets** → **+ New client secret**.
- 2 Give it a description (e.g. `cyprus-bot-secret`) and an expiry, then click **Add**.
- 3 **Immediately copy the Value** — it is shown only once. This is the bot password.
- 4 Add this value to `.env` as **MicrosoftAppPassword**.

Step 2 — Create the Azure Bot Resource

- 1 In the Azure portal, click **Create a resource** and search for **Azure Bot**.
- 2 Choose **Azure Bot** and click **Create**.
- 3 Set a **Bot handle** (e.g. `cyprus-ai-bot`).
- 4 For **Microsoft App ID**, select **Use existing app registration** and pick the app from Step 1.
- 5 Complete the wizard and create the resource.
- 6 Open the new bot resource, go to **Configuration**, and set the **Messaging endpoint** to:
`https://<your-domain>/api/messages`

If your domain is not live yet, leave it blank for now and set it after completing Step 5.

- 1 Open **Channels**, click the **Microsoft Teams** row, and choose **Configure** (or enable) the Teams channel to connect the bot to Teams.

Step 3 — Set the Bot Credentials in .env on Ubuntu

On the Ubuntu server, edit `/sml-genAI/.env` and confirm these values:

```
MicrosoftAppType=SingleTenant
MicrosoftAppId=<client-id-from-step-1>
MicrosoftAppPassword=<secret-value-from-step-1>
MicrosoftAppTenantId=<tenant-id-from-step-1>
```

Restart the application (e.g. `docker compose restart` or `bash scripts/start.sh`) so the new credentials are picked up.

Step 4 — Deploy and Expose the Endpoint over HTTPS

Microsoft Teams only accepts bot endpoints served over **valid HTTPS**. The application runs on port 8000; we use Nginx with a Let's Encrypt certificate in front of it. First, point a domain (or subdomain) at your Ubuntu server's public IP, then install Nginx and Certbot:

```
sudo apt update && sudo apt install -y nginx certbot python3-certbot-nginx
```

Create the Nginx site configuration:

```
sudo nano /etc/nginx/sites-available/cyprus

server {
    server_name <your-domain>;
    listen 443 ssl;
    ssl_certificate      /etc/letsencrypt/live/<your-domain>/fullchain.pem;
    ssl_certificate_key  /etc/letsencrypt/live/<your-domain>/privkey.pem;

    location /api/messages {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
    }
}

sudo ln -s /etc/nginx/sites-available/cyprus /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx

# Issue the SSL certificate
sudo certbot --nginx -d <your-domain>
```

Verify the endpoint responds (it should return an empty 200 OK to a bare POST, which confirms Nginx is forwarding correctly):

```
curl https://<your-domain>/api/messages
# or, after changing the firewall:
sudo ufw allow 'Nginx Full'
```

After HTTPS is live, return to **Step 2** and enter the final messaging endpoint in the Azure Bot resource.

Step 5 — Create the Teams App Package (manifest)

To let employees “Add” the bot, create a small Teams app package — a folder containing a `manifest.json` and two icon images. Create a directory and the manifest:

```
mkdir cyprus-ai-app && cd cyprus-ai-app
nano manifest.json
```

Paste this into `manifest.json` (replace the placeholders):

```
{
  "$schema": "https://developer.microsoft.com/en-us/json-schemas/teams/v1.17/MicrosoftTeams.schema.json",
  "manifestVersion": "1.17",
  "version": "1.0.0",
  "id": "<some-unique-guid>",
  "packageName": "com.sml.cyprusai",
```

```

"developer": {
  "name": "SML",
  "websiteUrl": "https://<your-domain>",
  "privacyUrl": "https://<your-domain>",
  "termsOfUseUrl": "https://<your-domain>"
},
"name": { "short": "Cyprus AI", "full": "Cyprus AI Policy Assistant" },
"description": {
  "short": "Company policy chatbot",
  "full": "Answers questions from company policy and procedure documents."
},
"icons": { "color": "color.png", "outline": "outline.png" },
"accentColor": "#FFFFFF",
"bots": [
  {
    "botId": "<MicrosoftAppId>",
    "scopes": ["personal", "team"],
    "supportsFiles": false,
    "isNotificationOnly": false
  }
],
"validDomains": [<your-domain>],
"permissions": ["identity", "messageTeamMembers"]
}

```

Add the required icon images to the same folder:

- **color.png** — 192x192 pixels, transparent background, full-colour logo.
- **outline.png** — 32x32 pixels, transparent background, white/transparent outline logo.

Zip the folder — the resulting `.zip` is the installable Teams app package:

```

cd ..
zip -r cyprus-ai.zip cyprus-ai-app

```

Step 6 — Publish and Let Employees Install

There are two ways to distribute the app, depending on whether you want to test first or roll it out company-wide:

6.1 Sideload (testing / small group)

- 1 In Teams, open the **Apps** store.
- 2 Click **Manage your apps** → **Upload an app** → **Upload a custom app**.
- 3 Select `cyprus-ai.zip` and click **Add**.

6.2 Company-wide via Teams Admin Center (production)

- 1 Go to <https://admin.teams.microsoft.com> with a Teams administrator account.
- 2 Open **Teams apps** → **Manage apps** → **Upload new app** and upload `cyprus-ai.zip`.
- 3 Submit/publish the app; define an app setup policy so it appears for the relevant employees.
- 4 Employees then go to **Apps**, search **Cyprus AI**, click **Add**, and use it in a 1:1 chat.

Important — User Access & Email Matching

The bot identifies an employee in Teams by their email or Azure AD object ID and looks them up in the PostgreSQL `users` table (see `app/bot/bot_handler.py`). For an employee to actually see answers, all of the following must be true:

- Their email in Teams must match the email in the **users** table exactly.

- They must exist in the **users** table (seeded by the admin portal or `scripts/init_db.py`).
- They must be assigned to a **group**, and that group must be linked to the **folder(s)** they should access.
- Folders must contain processed (READY) documents for them to receive answers.

If a user sees “you don't have access to any document folders”, it means one of the group/folder assignments is missing in the admin portal.

Quick Checklist

#	Task	Done
1	App registration created in Entra; client ID, tenant ID, secret saved	
2	Azure Bot resource created and connected to the app registration	
3	.env on Ubuntu updated with SingleTenant credentials	
4	Nginx + Let's Encrypt serving /api/messages over HTTPS	
5	Messaging endpoint set in Azure Bot resource	
6	Teams channel enabled on the bot	
7	Teams app package (manifest + icons) created and zipped	
8	App uploaded/published to the tenant	
9	Employee users + groups + folder access seeded in admin portal	
10	Live test in a Teams 1:1 chat	

Once the app is published and users are mapped correctly, employees simply open Teams, find Cyprus AI in the apps store, and add it to their chat to start asking policy questions.